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Research Article

In-Silico Molecular Docking Studies of Cymbopogon Nardus Compounds against Malarial Targets of Plasmodium falciparum

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ABSTRACT

Malaria caused by genus Plasmodium, is a parasite which is the main health issue for humans and about half of the population were suffered. Every year, approximately 1.2-2.7 million people died due to malaria globally. Therefore, to prevent the spreading of malaria from the glob novel active drugs with specific activities are necessary. The present study aimed to identify novel drug molecule together with the bioinformatic tools for the development of active malarial drugs. After brief study of physiological activity of *Plasmodium falciparum* few enzymes such as PfGST, PflDH and PfPKG are involved in the metabolism of the *P. falciparum* are observed as a target. Thirteen compounds which are components of *Cymbopogon nardus* used as ligands were got from the Pub Chem database and docked against the selected enzymes with Autodock software. Some of the compounds Germacrene-D-4-ol, Beta-cubebene, and Elemol showed drug-like characteristics and presents a significant antimalarial action in *in-silico* level. Results have outlined the significance of Compounds Germacrene-D-4-ol, Beta-cubebene, and Elemol as a promising antimalarial agent, which can be further evaluated and used for antimalarial drug development.

1. Introduction

Malaria is among the major vector-borne diseases that exact the heaviest toll from populations of sub-Saharan Africa. Worldwide, it is estimated that about 3.4 billion people are at risk of malaria [Murray CJL, 1996]. Within the past few years, approximately 207 million yearly cases of malaria occurred globally with the highest burden [Institute of Medicine (US), 2004] (80% of cases and 90% of deaths) recorded in Sub-Saharan Africa. The huge proportion of deaths (77%) occurs in children under 5 years [Breman JG, 2001]. The combined efforts of distributing long lasting insecticide treated mosquito nets, indoor residual sprays, and effective case management with potent antimalarials have dramatically declined the malaria related illness and mortality [Roger Ducos Fokouo Youmsi, 2017]. In India, nine Anopheline vectors are involved in transmitting malaria in diverse geo-ecological paradigms. About 2 million confirmed malaria cases and 1,000 deaths are reported annually, although 15 million cases and 20,000 deaths are estimated by WHO South East Asia Regional Office. India contributes 77% of the total malaria in Southeast Asia. Multi-organ involvement/dysfunction is reported in both *Plasmodium falciparum* and *P. vivax* cases. Most of the malaria

burden is borne by economically productive ages [Kumar A, 2007; Poojari et al., 2014].

India is one of the diverse countries in the aspect of habitat with all forms of land forms and diverse climatic conditions. Various types of plants growing around us were identified and utilized for various purposes. Utilization of plants for the medicinal use is seen since centuries in the name of Ayurveda [Chandra Prakash Kala, 2006]. Various insect repellent plants like *Ocimum suave*, *Ocimum kilimandscharicum*, *cymbopogon nardus*, *Melissa officinalis* and *Cymbopogon citratus* where studied. Out of all *cymbopogon nardus* is selected for further study as there was no study conducted on this plant in relation with *Plasmodium falciparum*. After considering various compounds retrieved from *Cymbopogon nardus*, Beta-cubebene, Beta-elemene, Limonene, Citronellal, Nerol, Citronellol, Geraniol, Citronellyl Acetate, Eugenol, Geranyl Acetate, Elemol, Neral and Germacrene-D-4-ol compounds have been selected for further evaluation and are used as ligands [Fatima EL Kamari, 2018].

Quinine, Chloroquine and Mepacrine were used as standard drugs against malaria in the past. Currently Artemisinin and its derivatives, Amodiaquine, Chloroquine derivatives like Hydroxychloroquine sulfate, Mefloquine (Structural analog of quinine) and Atovaquone are used as standard drugs [Belete

TM, 2020]. As the resistance for the current drugs towards malarial parasite species and development of new indigenous species have increased. So, there is a necessity to explore for new efficient drugs. Therefore, the objective of this study is to screen the phytochemicals of *C. nardus* against malarial targets through In-Silico Molecular docking.

2. Materials and Methods

2.1. Protein preparation

Among all the enzyme targets that involve in the Plasmodium falciparum three targets play important role in the life cycle of malarial parasite Plasmodium. Glutathione S-transferase (PfGST), Lactate dehydrogenase (PfLDH), and Cyclic GMP-dependent protein kinase (PfPKG) are used as targets. Hence, in this study these three targets were selected for molecular docking with selected ligands. The three-dimensional structure of Glutathione S-transferase (PfGST) (PDB: 1PA3), Lactate dehydrogenase (PfLDH) (PDB: 1LDG), and Cyclic GMP-

dependent protein kinase (PfPKG) (PDB: 5DZC) was downloaded from the RCSB protein Data Bank.

Using Discovery Studio Target proteins are prepared. The protein is selected and previously present ligand is removed from the protein. Then the hetero atoms and water molecules are deleted. The target protein is opened in Auto dock and hydrogens are added. Then added kollman charges and computer gasteiger. The prepared protein is saved in PDB format.

2.2. Ligands preparation

All the selected ligands are prepared in order to proceed for molecular docking. The ligands are prepared by using Auto dock. The structures of ligands are depicted in previous chapter. SDF format of files is converted to PDB files using OpenBabel (version 3.1.1) and then converted to PDBQT format using Auto dock to generate atomic coordinates. First the ligand is opened in Auto dock and then for this ligand choose root and

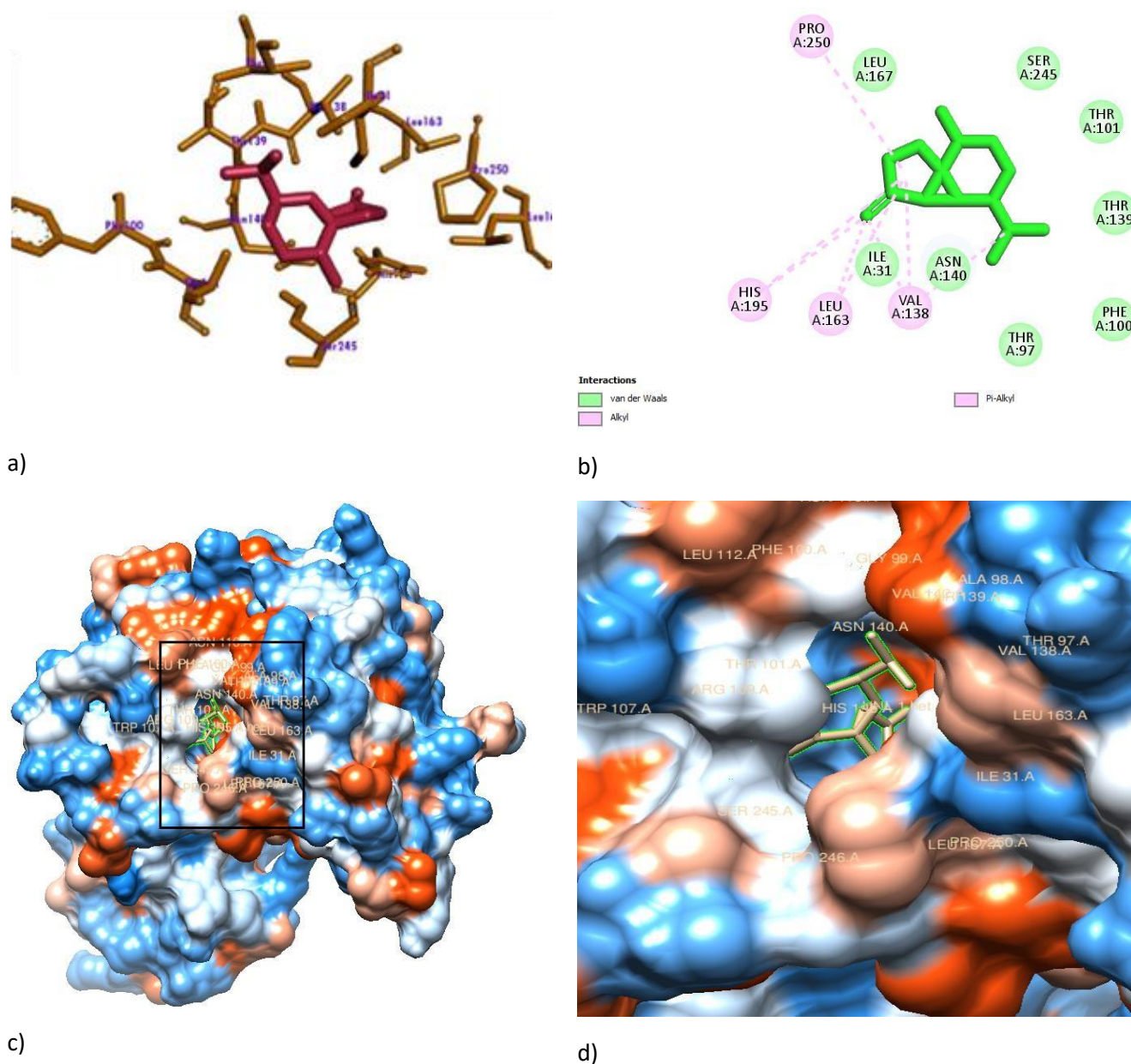
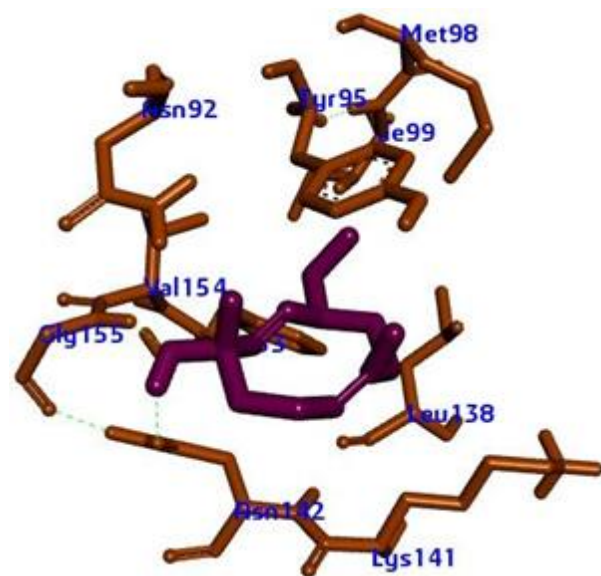
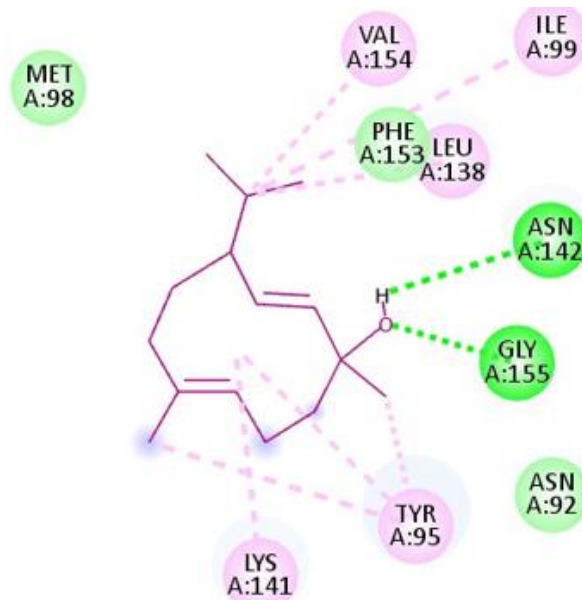


Fig. 1. Binding interactions of pLDH with phytochemicals. (a) 3D view interaction of Beta-cubebene with residues, (b) 2D interaction of Beta-cubebene with residues. (c) Surface interaction of Beta-cubebene (d) zoomed view of c



a)



b)

Fig. 2. Binding interactions of pFGST with phytochemicals (a) 3D view representation of Germacrene-D-4-ol with residues, (b) 2D representation of Germacrene-D-4-ol with residues

detect root was selected. Then accordingly number of torsions are set. This prepared ligand is saved in PDBQT format.

2.3. Preparation grid parameter file

Auto dock tool needs a pdbqt file to prepare the grid parameter file (gpf). In a new window to set the grid, click on Grid > Macromolecule > Open and open the target pdbqt file by macromolecule. Then click on Ligand > Input > Open > Choose Ligand > Select desired ligand. Grid > Macromolecule > Chose > Select protein. Similarly, click on Grid > Set map type > Open and open the ligand pdbqt file. Then select Grid > Grid box > Open and then set the grid map, grid size, as well as grid center in x, y, and z-direction. After that, the output file can be saved as a gpf file.

2.4. Preparation docking parameter file

For the preparation of the docking parameter file (dpf), click on Docking > Macromolecule > Set rigid filament > Open in the Auto dock window to open the target PDBQT. Similarly, ligand pdbqt can also be opened by clicking on Docking > Ligand > choose > Select ligand > Accept. Then, set the algorithm by clicking on Docking > Search Parameters > Genetic algorithm and setting docking parameters. Finally, click on Docking > Output > Lararckian GA and save it as a dpf file. Following this click Run > Run Auto grid > Select second browser > Auto grid file > open gpf file. Then select third browser > Grid gpf > Open. Then Auto dock is ready to run. Firstly, it runs. It required a proper time and needed the grid parameter file as well as a docking parameter file.

2.5. Analysis of target active binding sites

The active sites are the coordinates of the ligand in the original target protein grids, and these active binding sites of target

protein were analyzed using the Drug Discovery Studio and auto dock.

2.6. Molecular docking analysis

The molecular docking was done with the help of Autodock 4.2 to determine the biological interaction between the silver atoms and the protein targets. Autodock also offers to evaluate docking interactions and binding energies of a minimum of ten conformations along with a docking inhibition constant (K_a). By selecting Analyze > Docking > Open, you may view the findings by opening the dlg file. A popup will open, click "OK" and then further click Analyze > Conformation > Play > & > Show info.

3. Results and Discussion

The minimum binding energy indicated that the protein (target enzyme) was successfully docked with the ligand. In docking procedure, the three enzymes where simultaneously docked with the thirteen ligands. In this Lactate dehydrogenase (PFLDH) shown high affinity towards Beta- cubebene, Glutathione S-transferase (PFGST) show high affinity towards Germacrene-D-4-ol and cGMP-dependent protein kinase (PFPKG) show high affinity towards Beta-cubebene

3.1. Binding interactions of pFLDH with phytochemicals

The possible binding modes of the selected ligands with target proteins have been evaluated. The docking results showed that all the docking poses have the RMSD value of < 1.0 Å which indicated a valid docking pose. LDH protein residues Pro 250, His 195, Leu 163, Val 138, Leu 167, Ser 245, Thr 101, Thr 139, Phe 100, Thr 97, Asn 140, Ile 31 was formed interaction with Beta-cubebene ligand molecule. Beta-cubebene showed relatively good binding affinity which showed binding energy of -6.98 kcal/mol.

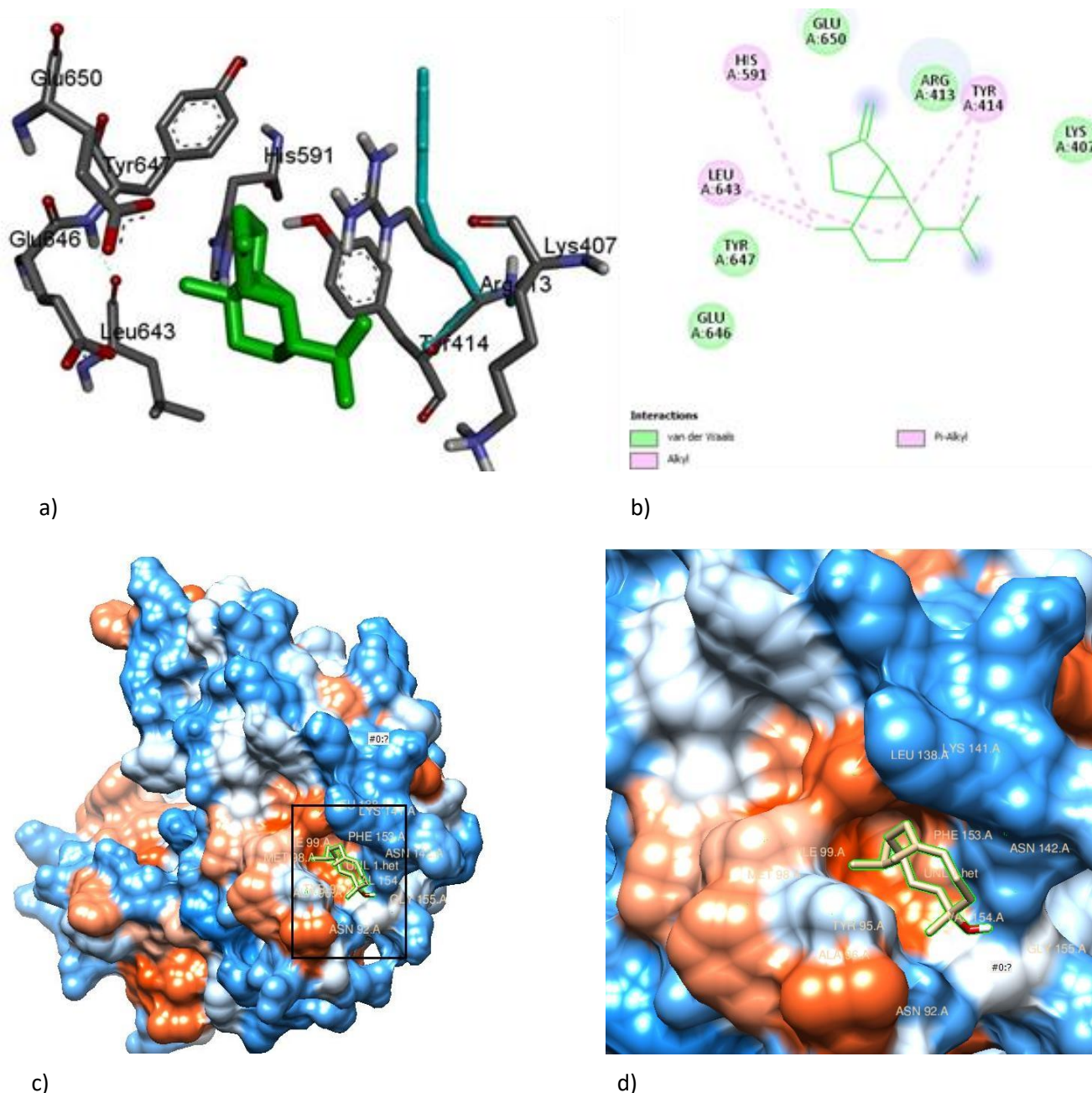


Fig. 3. Binding interactions of pfPKG with phytochemicals. (a) 3D view representation of Beta- cubebene with residues, (b) 2D representation of Beta-cubebene with residues. (c) Surface representation of Beta-cubebene (d) zoomed view of c

3.2. Binding interactions of pfGST with phytochemicals

In case of Glutathione S-transferase (PfGST) protein residues Asn 142 and Gly 155 formed Hydrogen bonds and Val 154, Ile 99, Leu 138, Tyr 95, Lys 141, Met 98, Phe 153 and Asn 92 formed hydrophobic interactions with Germacrene-D-4-ol ligand molecule. Germacrene-D-4-ol showed relatively high binding affinity which showed binding energy of -7.1 kcal/mol.

3.3. Binding interactions of pfPKG with phytochemicals

cGMP-dependent protein kinase (PfPKG) protein residues Glu 646, Tyr 647, Leu 643, His 591, Glu 650, Arg 413, Tyr 414 and Lys 407 has formed interaction with Beta-cubebene ligand molecule. Beta-cubebene showed relatively good binding affinity which showed binding energy of -5.92 kcal/mol.

4. Conclusion

Compounds Germacrene-D-4-ol, Beta-cubebene, and Elemol showed drug-like characteristics and presents a significant antimalarial action in *in-silico* level. Results have outlined the significance of Compounds Germacrene-D-4-ol, Beta-cubebene, and Elemol as a promising antimalarial agent, which can be further evaluated and used for antimalarial drug development.

Competing interests:

The authors declare that they have no competing interests

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S.No.	Name	CID	Target enzymes – Binding energy (Kcal/mol)		
			PfLDH (1LDG)	PfGST (1PA3)	PfPKG (5DZC)
1	Beta-cubebene	93081	-6.98	-6.53	-5.92
2	Beta-elemene	6918391	-6.44	-5.68	-4.87
3	Limonene	22311	-5.22	-4.88	-4.66
4	Citronellal	7794	-4.67	-4.87	-3.84
5	Nerol	643820	-4.96	-5.01	-3.89
6	Citronellol	8842	-4.57	-4.24	-3.49
7	Geraniol	637566	-4.44	-4.45	-4.23
8	Citronellyl Acetate	9017	-5.22	-4.27	-3.03
9	Eugenol	3314	-4.49	-4.94	-3.9
10	Geranyl Acetate	1549026	-5.43	-5.33	-4.21
11	Elemol	92138	-6.27	-6.12	-5.49
12	Neral	643779	-4.94	-5.1	-4.03
13	Germacrene-D-4-ol	5352847	-6.61	-7.1	-5.62
14	Chloroquine	2719	-5.85	-4.41	-3.86

Table 1. Interaction of *Cymbopogon Nardus* phytochemicals with respective Malarial Parasite target proteins

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